

Finite Element Method

Approximations of Trial Solutions for 2D and 3D Problems — Part 1
MECH4103 — Spring 2026

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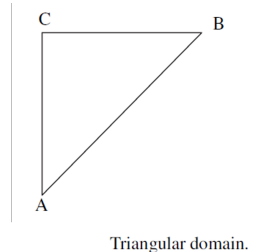
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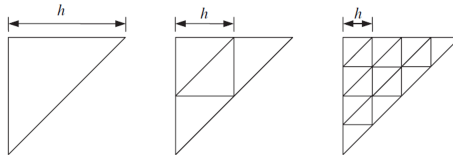
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Three-Node Triangular Element

- In this lecture, we describe the construction of the weight functions and trial solutions for multidimensional applications.
- We first consider the issue of **completeness**. To explain this concept, consider the simple problem domain shown in the figure.
- The domain is subdivided (meshed) into triangular elements, which is one of the finite elements to be considered in this lecture.



Triangular domain.



Finite element meshes of different refinements for the triangular domain.

Three-Node Triangular Element (cont.)

- The trial solution is then constructed on each element. Consider the following possible polynomial expansions:

$$(a) \quad \theta^e(x, y) = \alpha_1^e + \alpha_2^e x + \alpha_3^e y,$$

$$(b) \quad \theta^e(x, y) = \alpha_1^e + \alpha_2^e x + \alpha_3^e y^2,$$

$$(c) \quad \theta^e(x, y) = \alpha_1^e + \alpha_2^e x + \alpha_3^e y + \alpha_4^e xy + \alpha_5^e x^2 + \alpha_6^e x^3 y,$$

$$(d) \quad \theta^e(x, y) = \alpha_1^e + \alpha_2^e x + \alpha_3^e y + \alpha_4^e x^2 y^2 + \alpha_5^e xy + \alpha_6^e y^3.$$

$$(1) \quad \begin{array}{ccccccc} & & & & 1 & & \text{constant} \\ & & & & & & \\ & & x & & y & & \text{linear} \\ & & & & & & \\ x^2 & & xy & & y^2 & & \text{quadratic} \\ x^3 & x^2 y & & xy^2 & y^3 & & \text{cubic} \end{array}$$

Pascal triangle in 2D.

- Which one is a useful polynomial expansion for trial solutions? It can be determined by examining **Pascal's triangle**, which is shown in the figure.
- Each row of the triangle gives the monomials that must be included in a finite element approximation to provide an element with the order of completeness indicated to the right.
- If any of the terms in a row are missing, then the element will not be complete to that degree and will not have the convergence rate associated with that row of the expansion.
 - (1a) is *linear complete*, and its convergence rate will be of second order, i.e. quadratic.
 - (1b) will not converge, as it is missing the linear term in y .
 - (1c) and (1d) are not quadratically complete and will only have the convergence rate associated with a linear polynomial (quadratic in the displacements), even though they have monomials that are of higher order than linear.

- Complete polynomial expansions of linear, quadratic and cubic orders are given below:

linear: $\theta^e(x, y) = \alpha_0^e + \alpha_1^e x + \alpha_2^e y,$

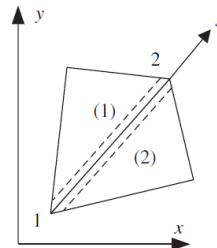
quadratic: $\theta^e(x, y) = \alpha_0^e + \alpha_1^e x + \alpha_2^e y + \alpha_3^e x^2 + \alpha_4^e xy + \alpha_5^e y^2,$

cubic: $\theta^e(x, y) = \alpha_0^e + \alpha_1^e x + \alpha_2^e y + \alpha_3^e x^2 + \alpha_4^e xy + \alpha_5^e y^2$
 $+ \alpha_6^e x^3 + \alpha_7^e x^2 y + \alpha_8^e y^2 x + \alpha_9^e y^3.$

- We next consider the issue of C^0 continuity. Consider the two adjacent elements shown in the figure, each with a complete linear polynomial expansion:

$$\theta^{(1)} = \alpha_0^{(1)} + \alpha_1^{(1)} x + \alpha_2^{(1)} y, \quad \theta^{(2)} = \alpha_0^{(2)} + \alpha_1^{(2)} x + \alpha_2^{(2)} y,$$

- where the superscripts indicate the element number. Each polynomial is obviously C^0 continuous within the element.



Continuity between two linear triangular elements.

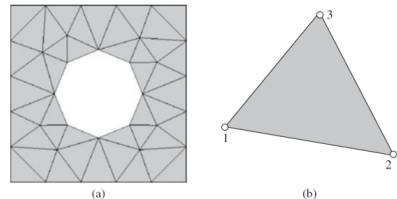
Continuity between two linear triangular elements.

- However, for the function to be globally C^0 , it must also be C^0 continuous at every point on the interfaces between the elements (not just at the nodes). In other words, for the specific example that we are considering, it is necessary that

$$\theta^{(1)}(s) = \theta^{(2)}(s).$$

Therefore, $\alpha_0^{(1)}$, $\alpha_1^{(1)}$, $\alpha_2^{(1)}$, $\alpha_0^{(2)}$, $\alpha_1^{(2)}$ and $\alpha_2^{(2)}$ have to be carefully chosen to satisfy C^0 continuity between the two elements.

- In the following, we describe how to construct continuous and complete shape functions for triangular elements.
- We will start with the **three-node triangular element**. The three-node triangular element is one of the most versatile and simplest of finite elements in two dimensions.
 - **Advantage:** One can easily represent almost any geometry with triangular elements.
 - **Disadvantage:** A disadvantage of the three-node triangle is that it is a relatively inaccurate element.
- Two finite element meshes consisting of three-node triangular elements are shown in the figure.



(a) Curved boundary approximation using three-node triangular finite elements and (b) a single three-node triangular finite element.

- The trial solution in each triangular element is approximated by a linear function of the spatial coordinates x and y :

$$\theta^e(x, y) = \alpha_0^e + \alpha_1^e x + \alpha_2^e y. \quad (2)$$

- The above can be written in the matrix form as shown below:

$$\theta^e(x, y) = \alpha_0^e + \alpha_1^e x + \alpha_2^e y = \underbrace{\begin{bmatrix} 1 & x & y \end{bmatrix}}_{\mathbf{p}(x, y)} \underbrace{\begin{bmatrix} \alpha_0^e \\ \alpha_1^e \\ \alpha_2^e \end{bmatrix}}_{\boldsymbol{\alpha}^e} = \mathbf{p}(x, y) \boldsymbol{\alpha}^e. \quad (3)$$

- We should be able to uniquely express the parameter α_i^e in terms of the nodal values θ_i^e :

$$\begin{aligned} \theta_1^e &= \alpha_0^e + \alpha_1^e x_1^e + \alpha_2^e y_1^e \\ \theta_2^e &= \alpha_0^e + \alpha_1^e x_2^e + \alpha_2^e y_2^e \\ \theta_3^e &= \alpha_0^e + \alpha_1^e x_3^e + \alpha_2^e y_3^e \end{aligned} \Rightarrow \underbrace{\begin{bmatrix} \theta_1^e \\ \theta_2^e \\ \theta_3^e \end{bmatrix}}_{\mathbf{d}^e} = \underbrace{\begin{bmatrix} 1 & x_1^e & y_1^e \\ 1 & x_2^e & y_2^e \\ 1 & x_3^e & y_3^e \end{bmatrix}}_{\mathbf{M}^e} \underbrace{\begin{bmatrix} \alpha_0^e \\ \alpha_1^e \\ \alpha_2^e \end{bmatrix}}_{\boldsymbol{\alpha}^e}. \quad (4)$$

- Taking the inverse of the above equation, we obtain an expression for the parameters in terms of the nodal values:

$$\boldsymbol{\alpha}^e = (\mathbf{M}^e)^{-1} \mathbf{d}^e. \quad (5)$$

- Substituting (5) into (3) gives

$$\theta^e(x, y) = \mathbf{p}(x, y) (\mathbf{M}^e)^{-1} \mathbf{d}^e. \quad (6)$$

- As in the one-dimensional case, the matrix product preceding $\mathbf{d}^e$ gives the shape functions:

$$\theta^e(x, y) = \mathbf{N}^e(x, y) \mathbf{d}^e. \quad (7)$$

- The shape functions are given by

$$\mathbf{N}^e(x, y) = \mathbf{p}(x, y)(\mathbf{M}^e)^{-1} \equiv [N_1^e(x, y) \quad N_2^e(x, y) \quad N_3^e(x, y)] \quad (8)$$

- To develop a closed form expression for the shape functions, it is necessary to invert the matrix $\mathbf{M}^e$, which gives

$$(\mathbf{M}^e)^{-1} = \frac{1}{2A^e} \begin{bmatrix} y_2^e - y_3^e & y_3^e - y_1^e & y_1^e - y_2^e \\ x_3^e - x_2^e & x_1^e - x_3^e & x_2^e - x_1^e \\ x_2^e y_3^e - x_3^e y_2^e & x_3^e y_1^e - x_1^e y_3^e & x_1^e y_2^e - x_2^e y_1^e \end{bmatrix}, \quad (9)$$

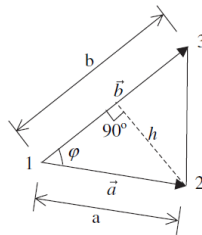
where A^e is the area of element e ; the determinant of $\mathbf{M}^e$ has been replaced by $2A^e$ and is given by

$$2A^e = \det(\mathbf{M}^e) = (x_2^e y_3^e - x_3^e y_2^e) - (x_1^e y_3^e - x_3^e y_1^e) + (x_1^e y_2^e - x_2^e y_1^e). \quad (10)$$

- Evaluating the above expression, we obtain

$$N_1^e = \frac{1}{2A^e} (x_2^e y_3^e - x_3^e y_2^e + (y_2^e - y_3^e)x + (x_3^e - x_2^e)y),$$

$$N_2^e = \frac{1}{2A^e} (x_3^e y_1^e - x_1^e y_3^e + (y_3^e - y_1^e)x + (x_1^e - x_3^e)y), \quad (11)$$

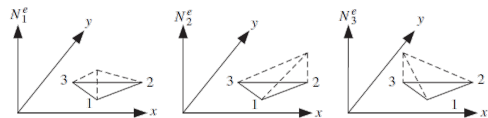


A diagram for computation of the area of a triangle.

A diagram for computation of the area of a triangle.

- You can easily check that if the nodes are numbered *clockwise*, Eq. (10) gives a *negative* area (as two rows of the determinant have been interchanged, which changes the sign). Therefore, **the nodes must be numbered counterclockwise**.
- The shape functions are drawn for a typical triangular element in the figure.
- It can be seen that each shape function vanishes at all nodes except one, and that node is the number on the shape function. In other words, these shape functions have the **Kronecker delta property**:

$$N_I^e(x_J^e, y_J^e) = \delta_{IJ}. \quad (12)$$



Three-node triangular element shape functions.

Three-node triangular element shape functions.

- We showed that the global shape functions $\mathbf{N}$ are given in terms of the element shape functions $\mathbf{N}^e$ by

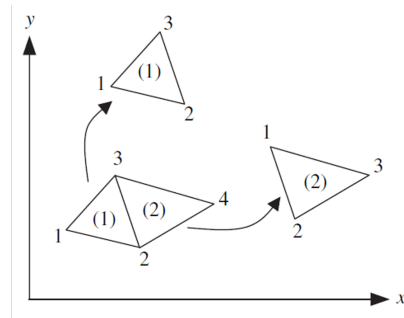
$$\mathbf{N}^T = \sum_{e=1}^{n_{el}} (\mathbf{L}^e)^T (\mathbf{N}^e)^T, \quad (13)$$

where $(\mathbf{L}^e)^T$ is the gather operator. The trial solutions are approximated by a linear combination of C^0 global shape functions:

$$\theta^h = \mathbf{N} \mathbf{d} = \sum_{I=1}^{n_{np}} N_I d_I, \quad (14)$$

so the same C^0 continuity of θ^h is guaranteed.

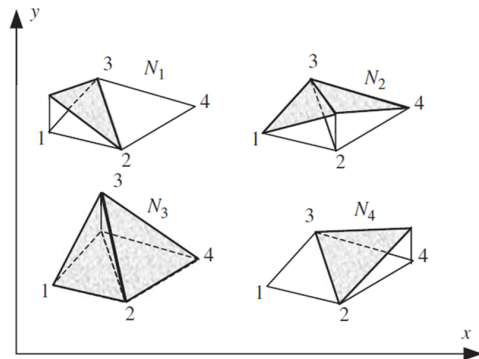
- For illustration, consider a two-element mesh shown in the figure.



Two-element mesh: local and global node numberings.

Two-element mesh: local and global node numberings.

- Therefore, the two element functions are equal along the interface and hence continuous across the interface.
- This argument holds for all other interfaces in the mesh, so the approximation is globally C^0 .
- Notice from the figure that the function has kinks along the interface, so the function is *not* C^1 .
- In a mesh of 3-node triangular elements, adjacent elements share two nodes on each interface, so global continuity is assured.



C^0 global shape functions for a two-element mesh. Only global node numbering is shown.

- The gradient of the shape functions matrix will be expressed in each element by a $\mathbf{B}^e$ matrix as before:

$$\nabla \theta^e = \begin{bmatrix} \frac{\partial \theta^e}{\partial x} \\ \frac{\partial \theta^e}{\partial y} \end{bmatrix} = \begin{bmatrix} \frac{\partial N_1^e}{\partial x} \theta_1^e + \frac{\partial N_2^e}{\partial x} \theta_2^e + \frac{\partial N_3^e}{\partial x} \theta_3^e \\ \frac{\partial N_1^e}{\partial y} \theta_1^e + \frac{\partial N_2^e}{\partial y} \theta_2^e + \frac{\partial N_3^e}{\partial y} \theta_3^e \end{bmatrix} = \underbrace{\begin{bmatrix} \frac{\partial N_1^e}{\partial x} & \frac{\partial N_2^e}{\partial x} & \frac{\partial N_3^e}{\partial x} \\ \frac{\partial N_1^e}{\partial y} & \frac{\partial N_2^e}{\partial y} & \frac{\partial N_3^e}{\partial y} \end{bmatrix}}_{\mathbf{B}^e} \underbrace{\begin{bmatrix} \theta_1^e \\ \theta_2^e \\ \theta_3^e \end{bmatrix}}_{\mathbf{d}^e} = \mathbf{B}^e \mathbf{d}^e.$$

- Referring to the shape functions (11) and the above, the $\mathbf{B}^e$ matrix is given by

$$\mathbf{B}^e = \frac{1}{2A^e} \begin{bmatrix} (y_2^e - y_3^e) & (y_3^e - y_1^e) & (y_1^e - y_2^e) \\ (x_3^e - x_2^e) & (x_1^e - x_3^e) & (x_2^e - x_1^e) \end{bmatrix}. \quad (15)$$

- Note that the $\mathbf{B}^e$ matrix is *constant* in each element, i.e. it is independent of x and y , and only depends on the coordinates of the nodes of the element.
- Thus, the gradient of any trial solution will be constant within any three-node triangular element.

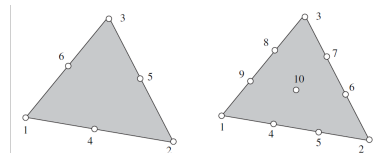
- From Pascal's triangle, a quadratic field in an element is given in terms of six parameters by

$$\theta^e(x, y) = \alpha_1^e + \alpha_2^e x + \alpha_3^e y + \alpha_4^e x^2 + \alpha_5^e xy + \alpha_6^e y^2. \quad (16)$$

- For continuity, the functions of two adjacent elements must have equal values at three points, and *three nodes* are needed along each edge.
- The approximation function for a cubic element is

$$\theta^e(x, y) = \alpha_0^e + \alpha_1^e x + \alpha_2^e y + \alpha_3^e x^2 + \alpha_4^e xy + \alpha_5^e y^2 + \alpha_6^e x^3 + \alpha_7^e x^2 y + \alpha_8^e y^2 x + \alpha_9^e y^3, \quad (17)$$

where *four nodes* are needed along each edge to ensure continuity. The nodal arrangement for the quadratic and cubic elements are shown in the figure.



(a) Six-node triangular finite element and (b) 10-node triangular element.

(a) Six-node triangular finite element and (b) 10-node triangular element.

- One difference between the quadratic and cubic elements is that the number of nodes on the edges is not equal to the number of constants.
- This imbalance is easily rectified by adding another node. It can be placed anywhere, but it is usually placed at the centroid.
- Elements of quartic order and higher can also be developed. However, elements of such high order are seldom developed from simple polynomial expansions.
- The drawback of these higher order triangular elements is that the resulting discrete system of equations are **not well conditioned**.
- Therefore, although such elements have potentially higher rates of convergence, and hence better accuracy, they are not used.

- The conditioning of a system is measured by its **condition number**, which is a ratio that reflects how much the solution might change in response to changes in the input data or coefficients of the system.
- For the stiffness matrix K in FEM, the condition number is typically defined as:

$$\kappa(K) = \frac{\text{largest eigenvalue of } K}{\text{smallest eigenvalue of } K}.$$

- Higher-order elements (e.g., quartic and beyond) involve larger polynomial degrees, which can lead to stiffness matrices with large condition numbers, making them “ill-conditioned.”
 - Ill-conditioned systems amplify errors during matrix operations, leading to inaccuracies and potential solver instability.
 - Ill-conditioned systems require more computational effort, as iterative solvers may converge very slowly or fail altogether without special techniques (e.g., preconditioning).
- Thus, while the higher-order triangular elements can theoretically offer better accuracy due to their improved convergence rates, their tendency to create ill-conditioned systems makes them less practical in most applications.

Due to round-off errors in element integration, assembly, and the linear solver, we effectively solve a perturbed system

$$(\mathbf{K} + \delta\mathbf{K})\hat{\mathbf{u}} = \mathbf{f} + \delta\mathbf{f}.$$

The relative error in the computed solution satisfies

$$\frac{\|\hat{\mathbf{u}} - \mathbf{u}\|}{\|\mathbf{u}\|} \leq \kappa(\mathbf{K}) \left(\frac{\|\delta\mathbf{K}\|}{\|\mathbf{K}\|} + \frac{\|\delta\mathbf{f}\|}{\|\mathbf{f}\|} \right),$$

showing that the condition number amplifies the effect of numerical perturbations.

Next: Approximations of Trial Solutions for 2D and 3D Problems — Part 2

Thanks for your attention!